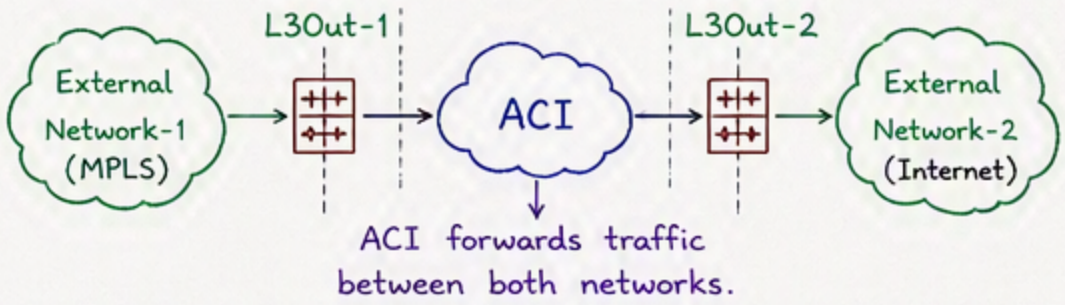
ACI NOTES - PART 6 TRANSIT ROUTING

① What is Transit Routing ?

ACI acts as a router between two external networks through two L3Outs.



② External EPGs

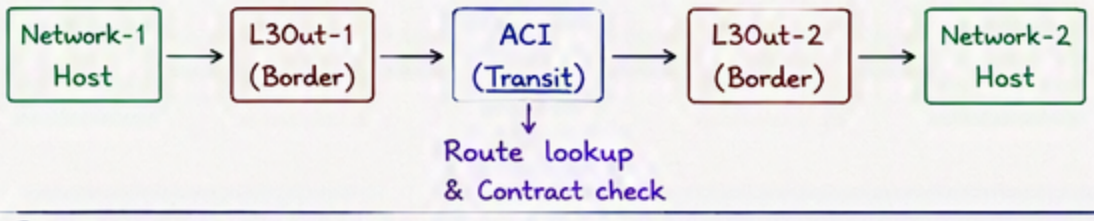
- External EPG represents external subnets.
- Created under L3Out.

External EPG	Subnets
epg_ext_mpls	10.100.0.0/16
epg_ext_internet	0.0.0.0/0

Note : Traffic matches destination IP with subnets configured in External EPG.

③ Traffic Flow (Network-1 → Network-2)

① Host in Network-1 sends traffic to Host in Network-2
② Traffic reaches L3Out-1 (Border Leaf)
③ Route lookup in ACI VRF
④ Destination matches External EPG (Ext-Net-2)
⑤ Contract check (Ext1 ↔ Ext2)
⑥ If Service Graph / PBR configured → Redirect
⑦ Traffic sent out via L3Out-2 to External Network-2
⑧ Return traffic follows the same path in reverse



④ Contracts in Transit Routing

Contracts are required between the two External EPGs.

Consumer : External EPG-1
Provider : External EPG-2
Filter : Define allowed protocols / ports

Contract must be applied in BOTH directions for bidirectional traffic.



⑤ Route Control Profiles (RCP)

A. Import Route Control

Controls what routes ACI accepts from outside into the fabric.



Example :
External router advertises :
- 0.0.0.0/0
- 10.0.0.0/8
- 172.16.0.0/16

Import Traffic Pollowt only : 0.0.0.0/0
Result : Only default route imported.

B. Export Route Control

Controls what routes ACI advertises to the external network.

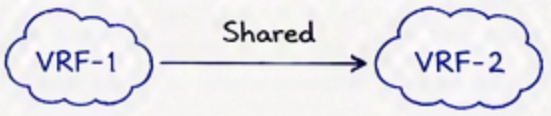


Example :
ACI has routes :
- 10.10.10.0/24
- 10.20.20.0/24
- 10.30.30.0/24

Export Policy Permit only : 10.10.10.0/24
Result : Only permitted routes advertised.

C. Shared Route Control

Used for route leaking between VRFs or between L3Outs.



Example :
Leak routes from VRF-SHARED to VRF-PROD using RCP.
- Only DNS subnet leaked
- Not all subnets.

⑥ External Subnet Flags (Under External EPG)

- Import Route Control Subnet → Route can be imported
- Export Route Control Subnet → Route can be exported
- Shared Route Control Subnet → Route used for leaking
- Regular Subnet → Only used for matching

Always apply correct flags to avoid route missing issues.

⑦ Route Advertisement Flow

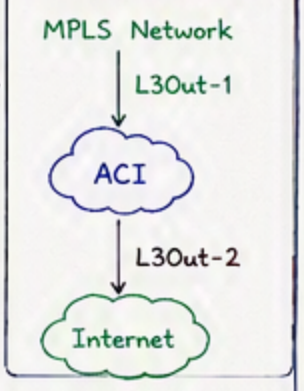
External Router → Border Leaf → MP-BGP EVPN → All Leafs

Advertises routes via BGP	Learns routes installs in VRF applies RCP	Advertises as Type-5 Route	Receive routes install in VRF

⑧ Important Considerations

- ✓ ACI does not modify packets (ho NAT).
- ✓ ACI only forwards/routes trafing (like a transit router).
- ✓ MTU must be consistent across L3Out links.
- ✓ Asymmetric routing can cause session issues.
- ✓ Proper route control and contracts are critical.
- ✓ Return path must be available, otherwise traffic will not flow back.

Real Life Example



⑨ Common Transit Routing Issues

- Route exists on Border Leaf but not on other Leafs → Check RCP / Type-5 Advertisement.
- Traffic forward works but return fails → Missing return route / contract.
- Wrong External EPG match → Check subnet configuration.
- Contract missing between External EPGs.
- Service Graph misconfigured.

⑩ Useful Verification Commands

show ip route vrf <vrf>	→ Check routes on leaf
show bgp l2vpn evpn summary	→ Check MP-BGP EVPN
show bgp l2vpn evpn route-type 5	→ Check Type-5 routes
show route-control profile <rcp>	→ Verify RCP configuration
show ip route vrf <vrf> <prefix>	→ Check specific route
show zoning-rule	→ Check contracts installed
show l3out	→ Check L3Out status
show tech-support	→ Complete troubleshooting

SUMMARY

1. Routes come in from outside → Import RCP
2. Routes go out to outside → Export RCP
3. Routes leak between VRFs/L3Outs → Shared RCP
4. Contracts control traffic between External EPGs
5. MP-BGP EVPN distributes Type-5 routes to all Leafs

Memory Aids

Import = Entering ACI
Export = Leaving ACI
Shared = Sharing inside
Type-2 = Endpoint (IP+MAC)
Type-5 = Network/Subnet

Final Rule

Route + Contract + Return Path
= Successful Communication